Measuring Mitosis

Student worksheet | Biology 2e, Chapter 10 | 40-55 minutes

Before you begin: using a microscope

A microscope uses light and lenses to reveal small structures. Begin with a low-power objective to locate the specimen, then use higher power and fine focus to inspect chromosome features. Total magnification equals eyepiece magnification multiplied by objective magnification.

The eyepiece is the lens you look through; this model has a 10x eyepiece. The objective is the lens closest to the slide. Multiply their magnifications: 10x eyepiece x 10x objective = 100x total; 10x eyepiece x 40x objective = 400x total. The cell images are original teaching illustrations, not microscope photographs. They are snapshots, not a movie of the cell cycle.

Observe and collect two fields

1. Record your prediction and sampling plan on the next page. Decide how to count each cell once and document uncertain classifications. Use the stage reference and consistent scoring rules.

2. In Collect observations, click Load prepared slide, then Turn light on. Start at the 10x objective. Adjust Fine focus until the cells are clear; adjust illumination if needed.

3. Switch to the 40x objective. Inspect the selected cell. If the view is unclear, adjust fine focus or illumination before choosing a stage.

4. Use the stage buttons directly below the microscope view to classify all 20 cells in Field A. Each choice records one cell and moves to the next unclassified cell. Use the numbered cell map to revisit a cell; changing its classification replaces the old choice without adding another cell.

5. Click Review field classifications. Compare with the practice key and revise if needed. Your own classifications are retained, including disagreements. Click Collect data and continue to save Field A and advance to Field B.

6. Repeat for Field B. Use the collected counts to calculate stage percentages and estimated durations. The assumed cycle duration is not a measured time.

After both fields are collected, choose Explore other settings. The exploration activity is at the end of this worksheet and uses a separate practice tally.

Measuring Mitosis

Student worksheet / prediction | Biology 2e, Chapter 10 | 40-55 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

What can a snapshot of dividing cells tell us about the cell cycle?

Load the prepared slide, adjust illumination and focus, and inspect cells at higher power. Classify and collect Field A, then Field B, before exploring microscope controls and cycle-duration assumptions separately. Use stage frequencies to estimate time in each stage under an explicitly assumed cycle duration.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: Use the same scoring definitions and count every cell in each selected field. Group prometaphase with prophase; count telophase/cytokinesis as one event until daughter cells are separated.

Select your route

Part	Virtual / Classroom	Equipment or conditions
Field A		
Field B		

Use the same questions for either route. Always label the source of each dataset.

Measuring Mitosis

Student worksheet / observations | Biology 2e, Chapter 10 | 40-55 minutes

2. Collect observations

Record a tally for two non-overlapping fields. The virtual fields contain 20 cells each; your classroom counts may differ. Group prometaphase with prophase.

Stage	Field A	Field B	Total	Percent	Est. hours
Interphase					
Prophase					
Metaphase					
Anaphase					
Telophase					
Total				100%	

Field A data source: _____ Field B data source: _____

Assumed cycle duration: _____ hours. State why this is an assumption:

Recognition notes

Identify two cells by field and number. Describe the chromosome features supporting each choice. Which stages were hard to distinguish, and why? What pattern do you notice across the cells? For classroom samples, also record exclusions.

Measuring Mitosis

Student worksheet / explanation | Biology 2e, Chapter 10 | 40-55 minutes

3. Explain the evidence

Estimated stage time = stage count / total count x assumed cycle duration. This assumes representative sampling of an asynchronous, steadily cycling population with comparable cycle lengths. Non-cycling cells, sampling bias, and differential cell loss can invalidate the estimate.

1. Which stage is most common? Give its count, percentage, and estimated duration, and distinguish observation from inference.

2. Describe the chromosome evidence you used to distinguish metaphase from anaphase. Why does interphase count even though it is not mitosis?

3. What assumptions are required to infer time from stage frequency? Explain why 24 hours is not a measured onion cell-cycle duration here.

4. Compare the two fields and identify one sampling limitation. Explain how you would improve the count.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.

Measuring Mitosis

Student worksheet / exploration | Biology 2e, Chapter 10 | 40-55 minutes

4. Explore after collecting both fields

Choose Explore other settings. Load the practice slide, turn on the light, focus at low power, and inspect selected cells at higher power. Explore illumination and focus. How do the controls affect your ability to distinguish chromosome features?

Classify a practice field and review it. Switching practice fields clears only the current practice tally. Your collected Field A and Field B data stay saved separately.

My question and prediction about microscope settings:

Settings tested and what I could observe:

Explore the time assumption

Keep the same practice tally. Compare estimated stage durations with assumed cycles of 12, 24, and 36 hours. The slider changes the estimates, not the cells or their counts. Explain why counts alone cannot measure elapsed time.

My comparison and explanation:

Practice estimates use the cells classified so far. Complete the whole field for a full-field tally. Practice settings and counts are included separately in exports.